NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Software Engineering

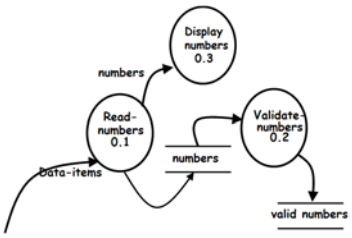
Assignment-6

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following can be said about the following DFD (data flow diagram)?



a. The bubble Display number processes numbers at the same rate as the bubble Read-numbers
b. The bubble Validate numbers processes numbers at the same rate as the bubble Read-numbers
c. The bubble Display number processes numbers at a different rate as compared to the bubble Read-numbers
d. The bubble Validate number processes numbers at a different rate as compared to the bubble Read-numbers
e. The rate at which the bubble Display numbers processes numbers is as the rate at which Validate numbers processes numbers.

Correct Answer: a. the bubble Display number processes numbers at the same rate as the bubble Read-numbers

d. The bubble Validate number processes numbers at a different rate as compared to the bubble Read-numbers

Detailed Solution:

If two bubbles are directly connected by a data flow arrow: They are synchronous

If two bubbles are connected via a data store: They are not synchronous.

So, among all the options, option **a.** and **d**. are correct.

QUESTION 2:

Which of the following are represented in the context diagram in a DFD (data flow diagram) model of a system?

a. The external entities with which the system interacts.
b. The data input to the system by the external entities
c. The important processes in the system
d. The output data generated by the system
e. The important data stores of the system

Correct Answer: a. The external entities with which the system interacts.
b. The data input to the system by the external entities
d. The output data generated by the system
Detailed Solution:

A context diagram shows: External entities, Data input to the system by the external entities, Output data generated by the system. It does not represent important process of the system and the important data stores. So, option **a., b.** and **d.** are the correct choices.

QUESTION 3:

Which of the following statements are not true of a data flow diagram (DFD) model?

a. It is a hierarchical diagram.
b. It represents code structure
c. It does not represent decisions and control flows.
d. It represents the decomposition of functions and data into finer forms.
e. It is constructed bottom-up

Correct Answer: b. It represents code structure
e. It is constructed bottom-up

Detailed Solution:
DFD is a hierarchical diagram it represents data flow among various process. It does not represent code structure of a program. Also, it does not represent decision and control flows. It is a top-down approach. So, option **b.** and **e.** are not true.

QUESTION 4:

Which of the following are **not** true of DFD (data flow diagram) modelling of a system?

a. Represents the system in multiple DFDs that are hierarchically arranged
b. The DFD model is usually constructed bottom-up
c. It explicitly represents the different processing carried out in the system
d. It explicitly represents data interchange among the different processes in the system.
e. The DFD model of a system can be directly implemented using a procedural language such as C.

Correct Answer: b. The DFD model is usually constructed bottom-up
e. The DFD model of a system can be directly implemented using a procedural language such as C.

Detailed Solution:

DFD model is usually constructed in top-down approach, DFD does not say in which programming language it can be implemented. So, option **b.** and **e.** are the correct choice.

QUESTION 5:

A context diagram should **not** represent which of the following?

a. External entities.
b. Data input to the system by the external entities,
c. Output data generated by the system.
d. Module structure
e. Order in which the external entities input data to the system

Correct Answer: d. Module structure
e. Order in which the external entities input data to the system

Detailed Solution:

A context diagram shows: External entities, Data input to the system by the external entities, Output data generated by the system.

So, option **d**. and **e**. are the correct options.

QUESTION 6:

Which of the following correctly characterizes balancing of DFDs (data flow diagrams)?

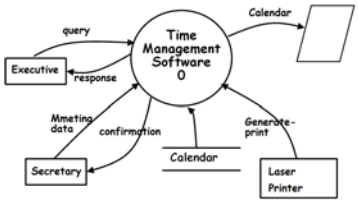
a. Data flowing into a bubble at any level must match the data flowing into the next level of DFD obtained by decomposing the bubble.
b. Data flowing out of a bubble at any level, other than level 0, must match the data flowing out of the previous level of DFD.
c. Data flowing out of a bubble at any level must match the data flowing into the next level of DFD obtained by decomposing the bubble.
d. Data flowing out of a bubble at any level must match the data flowing out of the next level of DFD obtained by decomposing the bubble.
e. Data flowing into a bubble at any level other than level 0 must match the data flowing into the previous level of DFD.

Correct Answer: a. Data flowing into a bubble at any level must match the data flowing into the next level of DFD obtained by decomposing the bubble
d. Data flowing out of a bubble at any level must match the data flowing out of the next level of DFD obtained by decomposing the bubble**.**

Detailed Solution:
Data flowing into or out of a bubble: Must match the data flows at the next level of DFD. Please refer slide no. 56 to 57 of week 6 lecture material.

QUESTION 7:

What are the mistakes in the following context diagram?



a. External entities (users) are appearing at the context level
b. Output occurring at the context level
c. Data item name missing on a data flow arrow
d. A data store appearing at the context level
e. Laser printer shown as an entity
f. Laser printer giving a command to the system

Correct Answer: d. A data store appearing at the context level
f. Laser printer giving a command to the system

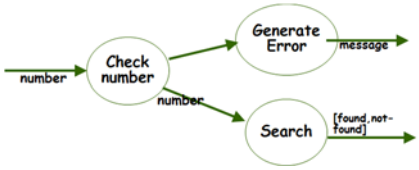
Detailed Solution:

A context diagram shows: External entities, Data input to the system by the external entities, Output data generated by the system.

So, option **d**. and **f.** are the correct options.

QUESTION 8:

What is wrong with the following DFD?



a. Unbalanced DFD
b. Data name missing on a data flow
c. Data name on data flowing out of the bubble “Search” is erroneous
d. Both data flow into and out of bubble “Check number” is “number”
e. Two different data cannot flow out of a single bubble

Correct Answer: b. Data name missing on a data flow

Detailed Solution:
In the given DFD, data name missing on the data flow 'Check number' to 'Generate Error'. So, option **b**. is correct.

QUESTION 9:

Which of the following types of information are not present in a structure chart model of a software system?

a. Various modules making up the system,
b. Module call structure
c. Data flow with external entities
d. Parameters passed among different modules
e. Data structures and algorithms of the modules

Correct Answer: c. Data flow with external entities
e. Data structures and algorithms of the modules

Detailed Solution:

A structure chart represents the following software architecture: Various modules making up the system, Module dependency (i.e., which module calls which other modules), Parameters passed among different modules. So, option c. and e. are the correct options.

QUESTION 10:

Which of the following statements about Data Flow Diagrams (DFDs) is correct?

a) DFDs are used to describe the physical layout of hardware components.

b) DFDs show the flow of control between different functions in a program.

c) DFDs represent the flow of data between processes, data stores, and external entities.

d) DFDs are only used during the implementation phase of the software development life cycle.

e) DFDs are primarily used for defining database schemas and relationships.

Correct Answer: c. DFDs represent the flow of data between processes, data stores, and external entities

Detailed Solution:
DFDs are used to model how data moves through a system — showing **what processes do**, **where data is stored**, and **how it interacts with external entities**.